MIL-S-17509(SHIPS)
26 February 1953

MILITARY SPECIFICATION

STEEL CASTINGS, AUSTENITIC, CHROMIUM-NICKEL, LOW MAGNETIC PERMEABILITY

1. SCOPE

1.1 Scope.- This specification covers austenitic chromium-nickel steel castings for use where low magnetic permeability is required and corrosion or elevated temperature may be encountered.

1.2 Classification.- Corrosion-resisting steel castings shall be of the following classes, as specified (see 6.2):

Class I - 18-9 Chrome-nickel.
Class II - 18-11 Chrome-nickel (Cb stabilized).
Class III - 18-10 Chrome-nickel.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL
NN-B-621 - Boxes, Wood, Nailed and Lock-Corner.

MILITARY
JAN-P-106 - Packaging and Packing for Overseas Shipment - Boxes, Wood, Nailed.
JAN-P-132 - Packaging and Packing for Overseas Shipment - Crates, Unsheathed, Wood, Nailed (for Maximum Net Load of 2,500 Pounds).
MIL-L-10547 - Liners, Case, Waterproof.
MIL-I-17214 - Indicator, Permeability; Low-MU (GO-NO GO).

GPO—O—NAV—H-351

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NAVY DEPARTMENT

General Specifications for Inspection of Material, and
Appendix II - Metals,
Part A - Definitions and Tests.
Part F - Radiography.
Section F-1 - Definitions and Radiographic Requirements.
46E4 - Electrodes, Welding (Covered), Corrosion Resisting (Austenitic Type) Steel.

STANDARDS

MILITARY

MIL-STD-23 - Nondestructive Testing Symbols.
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking of Shipments.

BUREAU OF SHIPS

Radiographic Standards for Steel Castings.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. - The material shall be such as to produce castings in full compliance with this specification.

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3.2 Chemical composition. - Chromium-nickel steel castings shall conform to the chemical requirements shown in table I.

Table I - Chemical composition.

Class	Carbon	Phosphorus (maximum)	Sulfur (maximum)	Chromium	Nickel	Manganese (maximum)	Silicon (maximum)	Columbium tantalum or titanium	Molybdenum (maximum)
	Percent	Percent	Percent	Percent	Percent	Percent	Percent		Percent
I	0.15 - 0.30	0.05	0.05	17.0 - 20.0	8.0 - 11.0	1.50	1.50	[1]	0.50
II	.10 max.	.05	.05	17.0 - 20.0	11.0 min.	1.50	1.50	[1]	.50
III	.08 max.	.05	.05	17.0 - 20.0	10.0 min.	1.50	1.50	-----	.50

[1] Columbium or columbium plus tantalum shall be not less than 10 times the carbon content and not more than 1.20 percent (tantalum shall not exceed 0.4 times the sum of the columbium and tantalum content).

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3.3 Mechanical properties. - Chromium-nickel steel castings shall conform to the mechanical properties shown in table II.

Table II - Mechanical properties.

Class	Tensile strength per square inch (minimum)	Yield point per square inch [1] (minimum)	Elongation in 2 inches (minimum)
	Pounds	Pounds	Percent
I	70,000	30,000	30
II	65,000	30,000	30
III	65,000	28,000	30

[1] The yield point shall be determined by means of dividers or at 0.05 percent offset.

3.4 Processing. - Unless otherwise specified in the contract or order, chromium-nickel steel castings shall be made by the electric furnace process.

3.5 Cleaning. -

3.5.1 Cleaning prior to inspection in the green. - Castings shall be well cleaned with heads and gates removed for Government inspection in such a manner as not to impair the contour of the casting. This shall be followed by such cutting or grinding as necessary to complete the removal of the heads or gates to the satisfaction of the Government inspector.

3.5.2 Cleaning after annealing. - Before final Government inspection, castings shall be thoroughly cleaned and all sand and scale removed by mechanical means for pickling. Blasting shall be followed by pickling if castings are to be used for corrosion resistance service (see 6.2).

3.5.3 Pickling after machining, or after blasting of castings intended for corrosion resistance service (see 6.2). - All castings shall be pickled as a final procedure after machining, drilling or cold work. This pickling shall consist of immersion of substantially scale free castings in a solution of nitric acid plus hydrofluoric acid for sufficient length of time to give a clean bright surface completely free of scale and foreign material after washing. Castings shall then be thoroughly washed and dried. If surfaces of the castings, which have been treated as specified above, are not bright and free of foreign matter, an additional treatment using nitric acid solution at elevated temperature shall be required. Castings shall then be thoroughly washed and dried.

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3.6 Internal chills and chaplets. - Internal chills shall not be used except by ific authorization by the bureau or agency concerned. When either chaplets or nal chills that remain a permanent part of the casting are used they shall be of a osition conforming to that specified for the casting in which they are employed.

3.7 Heat treatment. - All castings shall be annealed by heating uniformly in a erly constructed furnace at a temperature not less than 1, 50° Fahrenheit (F.) hour for each inch of thickness of the heaviest section, but in no case less than inutes, except that when satisfactory to the bureau or agency concerned, other oinations of time and temperature may be used. The castings shall be coole dly, whenever practicable, in water; otherwise, as adequate to insure quick ng of all parts through the carbide precipitation range. Before annealing, the ensions of the castings shall be sufficiently close to the finished dimensions, to re that the effect of the anneal and quench will reach the portions that will e sed by subsequent finish machining. In heavy section castings where considerable l may be removed by rough machining, a reheat treatment may be necessary, ss castings are rough machined before heat treatment.

3.8 Magnetic permeability. -

3.8.1 The magnetic permeability of the castings after heat treatment and cleaning, l not exceed 1.60 when tested with a permeability indicator (see 4.5.3.1). Any ing having a magnetic permeability in excess of 2.0 after machining shall be aced by the contractor.

3.8.2 When specified (see 6.2), the inherent magnetic permeability of the rior metal of each prototype casting shall be determined from a casting or from agnetic permeability test block cast for the purpose. Test specimens shall not ed a magnetic permeability of 1.3 at a field strength of 200 oersteds.

3.9 Radiographic examination. - Unless otherwise specified (see 6.2), radio- hic examination shall be performed on each casting in those locations on the ings indicated on suitable marked drawings (see Standard MIL-STD-23). The ernment inspector may require radiographic examination of additional locations nvestigate the extent of the defects revealed by surface inspection, machining, or ressure or other tests. In considering requests for authority to repair defects, ay, Gamma-ray or other nondestructive tests may be required to determine the nt of the defects and the effectiveness of the repairs. The acceptability of castings defects found by radiographic examination shall be judged by comparison with the iographic Standards for Steel Castings (see 6.3).

3.10 Resistance to intergranular corrosion (precipitated carbides). - Class III erial will be acceptable only if unsensitized specimens pass the embrittlement specified in 4.5.5. Material of class II will be acceptable only if sensitized cimens pass the embrittlement test specified in 4.5.5.

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3.11 Repair of defects. - Castings shall not be repaired, plugged, or welded without permission from the Government inspector, such permission will be given only for welding to be completed prior to final heat treatment and when the defects, after being thoroughly cleaned out to sound metal, are judged not to affect the strength, use or machinability of the castings. Welding of shrinkage cracks, or defects of a serious nature, will not be permitted unless satisfactory to the bureau or agency concerned. When welding is authorized, electrodes conforming to Specification 46E4 shall be used.

3.12 Marking. - The lot or heat number (indicating the date of casting) shall be cast or stamped on each casting unless a suitable method of identification by heats or lots satisfactory to the Government inspector is provided. Castings shall also be marked with the pattern or mark number (all raised letters).

3.13 Workmanship. - Castings shall be of uniform quality and condition, free from blowholes, porosity, hard spots, shrinkage defects, cracks or other injurious defects.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.2 Lot. -

4.2.1 For chemical analysis. - A lot shall consist of all castings from the same melt.

4.2.2 For other tests. - A lot shall consit of all castings of approximately the same maximum cross-sectional area from the same melt in one heat-treatment charge.

4.3 Sampling. -

4.3.1 For chemical analysis. - A sample for chemical analysis shall be taken by the Government inspector from each lot as specified in 4.2.1. Samples shall be obtained directly from the castings under inspection or from broken physical test specimens and shall be representative of the material. The drillings or millings shall be taken from sound metal below the surface, shall be fine, clean, and free from oil, dirt, grit, or foreign matter. The sample shall consist of not less than 2 ounces.

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3.2 For magnetic permeability test. - If specified, a test casting or a
tic permeability test block one foot long and having the maximum cross-
n of the casting it represents shall be cast from the same melt with the
ype casting from each pattern. This casting or block shall be separately
nd shall be heat-treated with the casting it represents. The permeability
nen shall be taken from the heaviest section of the casting or middle of the
(see 3.8.2 and 4.5.3).

3.3 For other tests. - At least one tension test specimen and for classes II and
intergranular corrosion test specimens shall be taken to represent each lot and
practicable, coupons for this purpose shall be attached to a casting. The number
cations of the specimens shall be such as to entirely exhibit the character of the
throughout the casting or castings represented, and the responsibility for
ding sufficient test coupons shall rest with the contractor. If it is impracticable
in sound castings with coupons attached, sufficient coupons shall be cast with
ted to the castings represented. If proper test coupons have not been provided,
uction casting may be selected by the Government inspector and specimens made
rom.

3.4 Sampling for inspection. - A random sample of castings shall be selected in
ance with table III, from each inspection lot of material offered for Government
tion of visual and surface characteristics with lot acceptance based on sampling
requirements in accordance with Standard MIL-STD-105.

Table III - Sampling for surface inspection
AQL (approx.) = .5 percent defective.

Number of castings in inspection lot	Number of castings in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 to 800	50	2	3
801 and over	75	3	4

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4.4 Inspection. - Unless otherwise specified (see 6.2), all sample castings shall be surface inspected in the green by the Government inspector and also inspected after annealing and pickling to determine compliance with requirements of this specification which do not require tests. Any casting in the sample that does not conform to this specification shall be rejected, and if the number of defective castings in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again for Government inspection provided the contractor has removed or repaired all nonconforming castings. The Government inspector shall again select and examine samples from such rejected lots to verify compliance with this specification.

4.5 Test procedures. -

4.5.1 Chemical analysis. - The sample obtained in accordance with 4.3.1 shall be forwarded to a Government laboratory designated by the bureau or agency concerned for chemical analysis. If the sample fails to conform to this specification all castings from that melt shall be rejected.

4.5.2 Tension tests. - Tension test specimens shall conform to the design and dimensions specified in General Specifications for Inspection of Material, Appendix II, Part A, for type I specimens.

4.5.3 Magnetic permeability tests. - When required, the magnetic permeability test specimen shall be machined from the casting or block specified in 4.3.2 to the following dimensions: 1/2 inch diameter by 3 inches long. It shall be forwarded to a laboratory designated by the bureau or agency concerned for test (see 3.8.2).

4.5.3.1 Each casting shall be tested on exposed surfaces with the permeability indicator conforming to Specification MIL-I-17214.

4.5.4 Radiographic examination. - Radiographic examination shall be performed under the supervision of the Government inspector. Radiographic examination shall be in accordance with General Specifications for Inspection of Material, Appendix II, Part F, Section F-1 and Radiographic Standards for Steel Castings.

4.5.5 Test for precipitated carbides (intergranular corrosion). -

4.5.5.1 Two specimens, each about 1/2 inch wide, approximately 1/4 inch thick, and approximately 3 inches long shall be prepared by sawing from a test bar or block representing each lot of casting submitted. The surfaces of the specimens shall be prepared by grinding or machining, or by filing with a 12-inch smooth flat mill file.

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4.5.5.2 Sensitizing treatment. -

(a) Class II intended for corrosion resistance service. - Specimens of class II material intended for corrosion resistance service shall be sensitized by heating within the range of 1,240 to 1,260° F. for 30 minutes.

(b) Class II intended for heat resistance service. - Specimens of class II material intended for heat resistance service shall be sensitized by heating within the range of 1,020 to 1,050° F. for 48 hours.

Specimens shall be cooled in air and pickled to remove any scale which may have formed.

4.5.5.3 Corrosion of specimens. - Nonsensitized specimens and class II specimens sensitized for 30 minutes at 1,250° F. shall be boiled 72 consecutive hours in a solution of the following composition: class II specimens sensitized for 48 hours at 1,035° F. shall be boiled 96 consecutive hours in a solution of the following composition:

Copper sulfate ($CuSO_4 \cdot 5H_2O$) --------	10 gm.
Sulfuric acid (sp. gr. 1.84) ----------	10 mℓ.
Water ----------	90 mℓ.

The specimens shall be covered with this solution using at least 30 mℓ of solution per square inch of specimens surface. A suitable flask shall be used fitted with a reflux condenser to prevent variations in concentration. After this treatment, the specimen shall be bent 150 degrees to a radius not greater than 1-1/2 times its thickness. The material shall be rejected if, after bending, cracks or other evidences of disintegration are visible to the naked eye or under low (30x) magnification.

4.5.6 Rejection. - If any test specimen is found to be not in conformance with this specification, the entire lot shall be rejected subject to the retest provisions of General Specifications for Inspection of Material, Appendix II, Part A. Any casting which exhibits high permeability shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging. -

5.1.1 Small polished casting. - Small polished castings weighing less than 50 pounds each shall be individually wrapped with a minimum of 2 thicknesses of 40 pounds minimum basis weight kraft paper or equivalent.

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5.1.2 Other castings. - Castings other than those specified in 5.1.1 will require no packaging.

5.2 Packing. -

5.2.1 For domestic shipment. -

5.2.1.1 Small rough castings. - Small rough castings weighing less than 150 pounds each shall be bundled or packed individually or in multiple units with adequate blocking or bracing in nailed wood boxes conforming to Specification NN-B-621. Bundles shall be metal strapped. The gross weight of nailed wood boxes shall not exceed 250 pounds and of bundles, 500 pounds.

5.2.1.2 Small polished castings. - Small polished castings weighing less than 150 pounds each including those packaged as specified in 5.1.1, shall be packed individually or in multiple units with adequate blocking or bracing in nailed wood boxes as specified in 5.2.1.1

5.2.1.3 Large castings. -

5.2.1.3.1 Large castings weighing more than 150 pounds each and having projections or surface finish which may be damaged in handling or shipping shall be packed individually or in multiple units with adequate blocking and bracing in an unsheathed crate conforming to Specification JAN-P-132 or in nailed wood boxes conforming to Specification NN-B-621. No strapping will be required. The castings shall be securely bolted, blocked, braced or strapped to prevent shifting or damage within the crate or box. The gross weight shall not exceed 500 pounds unless individual castings exceed this weight.

5.2.1.3.2 Large castings weighing more than 150 pounds each and not subject to damage in shipment may be shipped unpacked.

5.2.2 For overseas shipment. -

5.2.2.1 Small rough castings. - Small rough castings weighing less than 150 pounds each shall be bundled or packed as specified in 5.2.1.1, except that the gross weight of nailed wood boxes shall not exceed 150 pounds and of bundles, 250 pounds.

5.2.2.2 Small polished castings. - Small polished castings weighing less than 150 pounds each including those packaged as specified in 5.1.1, shall be packed as specified in 5.2.1.2, except that the gross weight shall not exceed 150 pounds. Shipping containers in which packaged castings are packed (see 5.1.1), shall be lined with a sealed waterproof case liner conforming to Specification MIL-L-10547 and appendix thereto.

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5.2.2.3 Large castings. - Large castings weighing more than 150 pounds shall be packed as specified in 5.2.1.3, except the nailed wood boxes shall conform to Specification JAN-P-106. Shipping containers shall be closed and strapped in accordance with the applicable container specification.

5.2.3 For domestic or overseas shipment. - Where practicable, shipping containers of similar construction shall be of uniform size, and shall contain the same number of castings of one size, melt or lot number. Containers shall be designed to fit the contents in a compact manner.

5.3 Marking. - In addition to any special marking required by the contract or order, each package and shipping container shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. - Castings are intended for use where austenitic corrosion resisting steel having low magnetic permeability is required. Class II castings may be used in steam applications at temperatures not exceeding 1050°F.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class required (see 1.2).
(c) Whether magnetic permeability test specimens are required (see 3.8.2).
(d) When radiographic examination is not required (see 3.9).
(e) Use of casting (see 3.5.2, 3.5.3, and 4.5.5).
(f) Whether inspection in the green is required (see 4.4).
(g) Whether domestic or overseas shipment is required (see 5.2).

6.3 Radiographic standards for steel castings, for the Navy are not at present available for general distribution. They may be examined at the following offices:

Inspectors of Naval Material.
Inspectors of Machinery, U.S.N.
Supervisors of Shipbuilding, U.S. N. and
Naval Inspectors of Ordnance.
Commanders, Naval Shipyards.

6.4 The packaging, packing and marking specified herein apply only to direct shipment to the Government and are not intended to apply to contracts or orders between the manufacturer and the prime contractor.

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6.5 Attention is invited to the necessity of supplying adequate information as to details of radiographic inspection to subcontractors, particularly regarding end use classification.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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